



# GUANGDONG NAN AO 26MW WIND POWER PROJECT

Document Prepared by

Timing Carbon Asset Management Co., Ltd.

## Contact Information

|                          |                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b>     | Guangdong Nan Ao 26MW Wind Power Project                                                                                                                 |
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| <b>Monitoring Period</b> | 29-September-2012 to 31-December-2017                                                                                                                    |
| <b>Prepared By</b>       | Timing Carbon Asset Management Co., Ltd.                                                                                                                 |
| <b>Contact</b>           | Room 155, 6/F Taohui Xintian, 219 Wangfujing Street, Dongcheng District, Beijing,<br>P.R.China<br><br>+86-13910964184<br><br>sammi.liu@timing-carbon.com |

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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Implementation Status of the Project

The purpose of the report is to calculate the emission reductions generated by Guangdong Nan Ao 26MW Wind Power Project (thereafter referred to the project) during the monitoring period (29/09/2012 to 31/12/2017), and to serve as basis for the verification and issuance of corresponding VCUs.

The Project, developed by China Resources Wind Power (Shantou) Co., Ltd, involves construction and operation of a wind power project that is sited on Nan Ao Island, Nan Ao County, Shantou City, Guangdong Province, People's Republic of China. The Project started construction on 10/06/2007 and was put into operation on 01/01/2008. The Project has been registered as a CDM project on 15/12/2008 (UNFCCC registration reference number: 1963).

The total installed capacity of the Project is 23.25MW consisting of 31 sets of wind turbine with unit capacity of 0.75MW. The electricity generated by the project is sold to the China Southern Power Grid (CSPG). For the local electric power supply prior to the start of the implementation of the project is from the CSPG dominated by grid-connected fossil fuel fired power plants, though replacing an equivalent amount of the grid electricity, the project, a zero-emission renewable wind power plant is expected to achieve GHG emission reductions, the annual amount of which is estimated up to 53,425 tCO<sub>2e</sub> over the first crediting period.

During this monitoring period (29/09/2012 to 31/12/2017), the project was implemented in line with the registered PD and the monitoring activities were conducted strictly in accordance with the revised monitoring plan approved by CDM EB. The Project has operated without any accidental or emergency events that might impact the accuracy and/or implementation of monitoring activities during this monitoring period. The net power supply during this monitoring period is 296,994.687 MWh. The total emission reductions claimed in this monitoring period (29/09/2012 to 31/12/2017) are 275,489 tCO<sub>2e</sub>.

## 1.2 Sectoral Scope and Project Type

Sectoral scope 1: Energy industries (renewable/non-renewable sources)

Project type: Grid-connected wind power project.

The project is not a grouped project.

## 1.3 Project Proponent

|                   |                                               |
|-------------------|-----------------------------------------------|
| Organization name | China Resources Wind Power (Shantou) Co., Ltd |
| Contact person    | Mr. Lin Weiping                               |

|                  |                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>     | Manager                                                                                                            |
| <b>Address</b>   | No. 127, Jinsha East Road, 14th floor Huaqiao Commercial Bank Building, Shantou, Guangdong province, P.R. of China |
| <b>Telephone</b> | +86-(0)754-88695785                                                                                                |
| <b>Email</b>     | linweiping@windpowerchina.com                                                                                      |

## 1.4 Other Entities Involved in the Project

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| <b>Organization name</b>   | Timing Carbon Asset Management Co., Ltd.                                                    |
| <b>Role in the Project</b> | Buyer                                                                                       |
| <b>Contact person</b>      | Liu Yanyan                                                                                  |
| <b>Title</b>               | Project Manager                                                                             |
| <b>Address</b>             | Room 155, 6/F Taohui Xintian, 219 Wangfujing Street, Dongcheng District, Beijing, P.R.China |
| <b>Telephone</b>           | +86-13910964184                                                                             |
| <b>Email</b>               | sammi.liu@timing-carbon.com                                                                 |

## 1.5 Project Start Date

01/01/2008, on which the VCS project began reducing GHG emissions.

## 1.6 Project Crediting Period

01/01/2008 - 31/12/2017 with a total number of 10 years.

There is a deviation for the crediting period. The project is registered under VCS Standard 2007.1 and completed validation before 19/03/2020, thus it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project shall be updated from 01/01/2008-14/12/2008 to 01/01/2008 - 31/12/2017. Since the renewal of the crediting period has not been completed within two years after the end of the first crediting period, the total VCS crediting period ends on 31/12/2017.

## 1.7 Project Location

The Project is located on Nan Ao Island, Nan Ao County, Shantou City, Guangdong Province, People's Republic of China, and the central of geographical coordinates are east longitude of 117°5'27.73"and north latitude of 23°25'12.63".

Geographical coordinates of each wind turbine of the project are listed below:

**Table 1: Geographical coordinates of each wind turbine**

| No. | Type    | Longitude (E)   | Latitude (N)   |
|-----|---------|-----------------|----------------|
| 1.  | S48/750 | 117° 05'17.33'' | 23° 25'53.03'' |
| 2.  | S48/750 | 117° 05'22.73'' | 23° 25'51.15'' |
| 3.  | S48/750 | 117° 05'30.16'' | 23° 25'52.33'' |
| 4.  | S48/750 | 117° 05'33.37'' | 23° 25'50.31'' |
| 5.  | S48/750 | 117° 05'38.42'' | 23° 25'46.64'' |
| 6.  | S48/750 | 117° 05'44.08'' | 23° 25'43.10'' |
| 7.  | S48/750 | 117° 05'44.62'' | 23° 25'38.53'' |
| 8.  | S48/750 | 117° 04'59.54'' | 23° 25'45.17'' |
| 9.  | S48/750 | 117° 05'07.80'' | 23° 25'28.97'' |
| 10. | S48/750 | 117° 05'10.77'' | 23° 25'36.39'' |
| 11. | S48/750 | 117° 05'11.00'' | 23° 25'40.07'' |
| 12. | S48/750 | 117° 05'17.58'' | 23° 25'41.33'' |
| 13. | S48/750 | 117° 05'18.52'' | 23° 25'34.77'' |
| 14. | S48/750 | 117° 05'20.99'' | 23° 25'30.56'' |
| 15. | S48/750 | 117° 05'26.98'' | 23° 25'32.11'' |
| 16. | S48/750 | 117° 05'27.94'' | 23° 25'28.46'' |
| 17. | S48/750 | 117° 05'33.06'' | 23° 25'28.45'' |
| 18. | S48/750 | 117° 05'18.85'' | 23° 25'07.72'' |
| 19. | S48/750 | 117° 05'17.10'' | 23° 25'04.16'' |
| 20. | S48/750 | 117° 05'27.73'' | 23° 25'10.06'' |
| 21. | S48/750 | 117° 05'29.31'' | 23° 25'07.82'' |
| 22. | S48/750 | 117° 05'33.41'' | 23° 25'07.95'' |

|     |         |                 |                |
|-----|---------|-----------------|----------------|
| 23. | S48/750 | 117° 05'35.09'' | 23° 25'01.21'' |
| 24. | S48/750 | 117° 04'53.94'' | 23° 25'49.08'' |
| 25. | S48/750 | 117° 04'20.53'' | 23° 26'16.05'' |
| 26. | S48/750 | 117° 04'23.10'' | 23° 26'13.16'' |
| 27. | S48/750 | 117° 05'14.49'' | 23° 25'28.57'' |
| 28. | S48/750 | 117° 05'13.87'' | 23° 25'22.00'' |
| 29. | S48/750 | 117° 05'16.00'' | 23° 25'16.84'' |
| 30. | S48/750 | 117° 05'17.99'' | 23° 25'14.11'' |
| 31. | S48/750 | 117° 05'24.93'' | 23° 25'15.00'' |

## 1.8 Title and Reference of Methodology

Approved consolidated baseline and monitoring methodology:

ACM0002 “Consolidated methodology for grid-connected electricity generation from renewable sources”, version 06

- The methodology also refers to the approved versions for the following tools:
- Tool for the demonstration and assessment of additionality, version 03;

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/approved.html>

<https://cdm.unfccc.int/Reference/tools/index.html>

## 1.9 Participation under other GHG Programs

The Project was registered as a CDM project on 15/12/2008 (Ref. 1963). The first CDM crediting period is from 15/12/2008 to 14/12/2015. CERs of 193,689 tCO<sub>2</sub>e have been issued for the monitoring period from 15/12/2008 to 28/09/2012. The emission reductions during this monitoring period (29/09/2012 to 31/12/2017) will only apply for issuance under VCS, which is ensured by the statement that the PP will not request the issuance of CERs under CDM and the VCU will not be double counted.

## 1.10 Other Forms of Credit

### Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-

ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO<sub>2</sub>e/year, not including renewable project<sup>1</sup>.

Thus, the project proponent: China Resources Wind Power (Shantou) Co., Ltd. as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Moreover, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS.

Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms. Furthermore, a statement on no double counting will be submitted to Verra to confirm the credits during this monitoring period has not been counted and will not be counted under emission trading programs and other binding limits.

#### Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

### 1.11 Sustainable Development Contributions

The Project activity will not only supply renewable electricity to grid, but also contribute to sustainable development of the local community, which mainly includes the following:

- The project utilizes wind resources to generate and supplied 296,994.687MWh renewable electricity to the power grid during this monitoring period, which contributes to SDG 7 and corresponding to the target of increasing the share of clean energy consumption of China's "Action Plan for Carbon Dioxide Peaking Before 2030".
- The project provides 11~12 long-term job opportunities for local residents during this monitoring period, which has a positive effect on the local economy which contributes to SDG 8, corresponding to China's target of "Increase on labour force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment".
- The project utilizes zero-emission wind power to supply electricity to the grid and reduces 275,489 tCO<sub>2</sub>e of GHG emissions during this monitoring period, which contributes to SDG13. corresponding to China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry,

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<sup>1</sup> [http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330\\_826728.html](http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html)

water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas.

For evidence of SDGs, please refer to Appendix 1 for details.

Table 2: Sustainable Development Contributions

| Row number | SDG Target | SDG Indicator                                                                                     | Net Impact on SDG Indicator        | Current Project Contributions                                                                                                                                                                                                                                                                   | Contributions Over Project Lifetime <sup>2</sup>                                                                                    |
|------------|------------|---------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1)         | 7.2        | 7.2.1 Renewable energy share in the total final energy consumption                                | Implemented activities to increase | The project has provided 296,994.687 MWh renewable energy to the grid during this monitoring period.                                                                                                                                                                                            | The project has provided 296,994.687 MWh renewable energy generation accumulated at the end of this monitoring period.              |
| 2)         | 8.5        | 8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities | Implemented activities to increase | The project has employed 11~12 persons including 8~9 men and 3 women during this monitoring period with yearly average salary: from 72,000RMB in 2012 to 123,480 RMB in 2017 which are about 138.2%~201.4% higher than the local average level (from CNY 30,226 in 2012 to CNY 40,975 in 2017). | 11~12 people including 3 females have been employed by the project as long-term employee by end of the reporting period.            |
| 3)         | 13.0       | Tonnes of greenhouse gas emissions avoided or removed                                             | Implemented activities to increase | By supplying 296,994.687 MWh renewable energy to the grid, the project has prevented the release of 275,489 tonnes of carbon into the atmosphere during the monitoring period.                                                                                                                  | The project prevented the release of 275,489 tonnes of carbon into the atmosphere accumulated at the end of this monitoring period. |

<sup>2</sup> This is the first time to report sustainable development contributions of the project. Therefore, the information for the project lifetime and the current monitoring period are the same as per VCS requirement.

## 2 SAFEGUARDS

### 2.1 No Net Harm

The environmental impact assessment (EIA) of the project has been approved by Shantou Environmental Protection Bureau on 13/08/2004. The EIA has assessed every possible aspect of environmental impacts of the project and recommended corresponding measures, where applicable. The environmental impacts and corresponding mitigation measures during operation have been discussed in the registered CDM-PDD. No negative environmental impacts have been identified.

Furthermore, the project makes positive contribute to the sustainable development as described in section 1.11 of this report e.g., providing job opportunities and clean energy to the local community, and mitigating GHG emissions.

In conclusion, construction and operation of the project does not cause any negative environmental nor socio-economic impacts.

### 2.2 Local Stakeholder Consultation

#### **LSC prior to the project implementation**

Local stakeholder consultation questionnaire survey has been conducted in April 2007. All comments received have been well addressed prior to the project implementation. Please refer to section E of the registered CDM-PDD of the project for detailed information. In conclusion, there was no significant negative comments received in the project preparation stage.

#### **LSC during the operation period**

During this monitoring period, the project carried out the communication with local stakeholders in line with the on-going communication mechanism, i.e..

The project owner published the contact information of the contact person who is responsible for stakeholders' comments to the local government and residents. Stakeholders were informed of the contact information, and their comments can be directly collected by the contact person. The comments would be fed back to the stakeholders by the contact person for a timely response. Besides, the contact person of project owner also meets local villagers to collect their comments and suggestions at least yearly i.e. in October 2012, November 2013, October 2014, November 2015, November 2016, and October 2017 during this monitoring period. Once the contact person received negative comments from the stakeholders, the contact person would record the negative comments and the feedback. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations.

In line with VCS requirements all the processed have been implemented to receive comments from local stakeholders as well as communicate with them. By the end of this monitoring period, the project did not receive any negative comments nor grievance from the stakeholders.

## 2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, and this section is not required.

# 3 IMPLEMENTATION STATUS

## 3.1 Implementation Status of the Project Activity

The total installed capacity of the Project is 23.25MW consisting of 31 sets of wind turbine with unit capacity of 0.75MW. Wind turbines and their corresponding generators employed by the Project were provided by Goldwind Science & Technology Co., Ltd, with model of Goldwind S48/750. Please see Table 2 below for key technical specifications.

Table 3. Major technical parameters of the key equipments of the Project

| Item                       | Unit           | Index                              |
|----------------------------|----------------|------------------------------------|
| Rated capacity             | kW             | 750                                |
| Number of blades           | /              | 3                                  |
| Rotor diameter             | m              | 44                                 |
| Swept area                 | m <sup>2</sup> | 2124                               |
| Startup speed              | m/s            | 4                                  |
| Rated wind speed           | m/s            | 16                                 |
| Shutoff speed              | m/s            | 25                                 |
| Rated voltage of generator | V              | 690                                |
| Power conditioner          | /              | Pitch regulated and variable speed |
| Control system             | /              | PC                                 |

No abnormal circumstance occurred during this monitoring period. There is no event or situation occurred during the monitoring period, which may impact the applicability of the methodology and may impact the GHG emission reductions and monitoring. The project was operational as normal during the monitoring period.

## 3.2 Deviations

### 3.2.1 Methodology Deviations

No methodology deviation exists.

### 3.2.2 Project Description Deviations

There is a deviation for the crediting period. The project is registered under VCS Standard 2007.1 and completed validation before 19/03/2020, thus it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project shall be updated from 01/01/2008-14/12/2008 to 01/01/2008 - 31/12/2017. Since the renewal of the

crediting period has not been completed within two years after the end of the first crediting period, the total VCS crediting period ends on 31/12/2017.

The deviation on the crediting period has no impact on the applicability of the methodology, additionality of the project nor the appropriateness of the baseline scenario.

There was a revised monitoring plan submitted, which has been approved by CDM Executive Board on 09/07/2010. After the project started operation, another project, Nanguo Wind Farm installed eight new turbines with each capacity of 750 kW next to the project site which started operation since Nov. 2008 and shares the same meters (M1 and M3) installed at the 110kV local substation to monitor electricity generated by both projects. Thus, the monitoring diagram, monitoring parameter and the formula to determine the net electricity exported by the project have been revised to deduct the electricity generation of the Nanguo Wind Farm. During this monitoring period, the monitoring activity was carried out in compliance with the revised monitoring plan. This revision does not impact the applicability of the methodology, additionality of the project nor the appropriateness of the baseline scenario. Details of the data and parameters monitored please refer to Section 4 below.

Reference for the approved revised monitoring plan:

<http://cdm.unfccc.int/UserManagement/FileStorage/L6Z4QVPONB1KFH2W7J38YRA9ITDXSU>

### 3.3 Grouped Projects

The Project is not a grouped project.

## 4 DATA AND PARAMETERS

### 4.1 Data and Parameters Available at Validation

| Data / Parameter                                                                             | EF <sub>y</sub>                                                                                          |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Data unit                                                                                    | tCO <sub>2</sub> e/MWh                                                                                   |
| Description                                                                                  | The combined margin grid emission factor of the Southern China Power Grid where the Project connected to |
| Source of data                                                                               | Registered CDM-PDD                                                                                       |
| Value applied                                                                                | 0.9276                                                                                                   |
| Justification of choice of data or description of measurement methods and procedures applied | Determined ex-ante and fixed for the 1 <sup>st</sup> crediting period                                    |
| Purpose of Data                                                                              | Calculation of baseline emissions.                                                                       |

|          |                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comments | Other parameters fixed ex-ante (i.e. $EF_{OM}$ , $EF_{BM}$ , $F_{i,j,y}$ , $NCV_{i,y}$ , $EF_{CO2,i,y}$ , Carbon Oxidation Factor, Installed Capacity, Electricity Generation and Electricity self-consumption ratio) were used to calculate the combined margin grid emission factor ( $EF_y$ ) during validation stage. |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4.2 Data and Parameters Monitored

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | $EG_{output,y}$                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Data unit                                                       | MWh                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description                                                     | Electricity supplied by the Project and the Nanguo wind farm during year y                                                                                                                                                                                                                                                                                                                                                                                  |
| Source of data                                                  | Readings of electricity meter M1                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description of measurement methods and procedures to be applied | Electricity supplied to the grid by the Project and Nanguo Wind Farm is monitored through the bidirectional electricity energy meters main meter M1 and its backup meter M3 operated by PP. The main meter M1 is installed at the outlet of 110kV local substation at the project site. Meter M3 is installed at the inlet of the 110kV local substation as the backup meter in case the error happens to the main meter (M1).                              |
| Frequency of monitoring/recording                               | Continuously measured and monthly recorded                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Value monitored                                                 | 371,408.087                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Monitoring equipment                                            | See table 3 below for details                                                                                                                                                                                                                                                                                                                                                                                                                               |
| QA/QC procedures to be applied                                  | Main meter M1 with an accuracy degree of 0.5S and backup meter M3 with an accuracy degree of 0.5S have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448), and compliance with the requirement of revised monitoring plan. Meter readings are crosschecked with ETNs. All data and records to be achieved for at least 2 years after the end of last crediting period. |
| Purpose of the data                                             | Calculation of baseline emissions                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Calculation method                                              | -                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Comments                                                        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                  |                                                                                           |
|------------------|-------------------------------------------------------------------------------------------|
| Data / Parameter | $EG_{import,y}$                                                                           |
| Data unit        | MWh                                                                                       |
| Description      | Electricity purchased from the Grid by the Project and the Nanguo wind farm during year y |
| Source of data   | Readings of electricity meter M1                                                          |

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description of measurement methods and procedures to be applied</b> | Electricity imported from the grid by the Project and Nanguo Wind Farm is monitored through the bidirectional electricity energy meters main meter M1 and its backup meter M3. The main meter M1 is installed at outlet of the 110kV local substation at the site. Meter M3 is installed at the inlet of the 110kV local substation as the backup meter in case the error happens to the main meter (M1).                                                   |
| <b>Frequency of monitoring/recording</b>                               | Continuously measured and monthly recorded                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Value monitored</b>                                                 | 1,113.200                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Monitoring equipment</b>                                            | See table 3 below for details                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>QA/QC procedures to be applied</b>                                  | Main meter M1 with an accuracy degree of 0.5S and backup meter M2 with an accuracy degree of 0.5S have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448), and compliance with the requirement of revised monitoring plan. Meter readings are crosschecked with ETNs. All data and records to be achieved for at least 2 years after the end of last crediting period. |
| <b>Purpose of the data</b>                                             | Calculation of baseline emissions                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Calculation method</b>                                              | -                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Comments</b>                                                        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data / Parameter</b>                                                | <b>EG<sub>Nanguooutput,y</sub></b>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Data unit</b>                                                       | MWh                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Description</b>                                                     | Electricity generated by Nanguo wind farm during year y                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Source of data</b>                                                  | Readings of electricity meters M2,1-M2,8                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Description of measurement methods and procedures to be applied</b> | There are eight main meters (M2,1-M2,8) installed at the 8 wind turbines of Nanguo Wind Farm which monitors the total electricity generated by Nanguo wind farm. To ensure the electricity supplied by Nanguo Wind Farm correctly measured and recorded in case that the main meters (M2,1-M2,8) are damaged, there are eight backup meters (M4,1-M4,8) installed at the 8 wind turbines as well to monitor the electricity generated by Nanguo wind farm. |
| <b>Frequency of monitoring/recording</b>                               | Continuously measured and monthly recorded                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Value monitored</b>                                                 | 733,00.200                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Monitoring equipment</b>                                            | See table 3 below for details                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>QA/QC procedures to be applied</b>                                  | Meters M2,1-M2,8 (all with accuracy of 0.2S) and M4,1-M4,8 (all with accuracy of 0.5S) have been properly calibrated annually according to                                                                                                                                                                                                                                                                                                                 |

|                            |                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | the requirement from Technical administrative code of electric energy metering (DL/T448), and compliance with the requirement of revised monitoring plan. All data and records to be achieved for at least 2 years after the end of last crediting period. |
| <b>Purpose of the data</b> | Calculation of baseline emissions                                                                                                                                                                                                                          |
| <b>Calculation method</b>  | -                                                                                                                                                                                                                                                          |
| <b>Comments</b>            | -                                                                                                                                                                                                                                                          |

Table 4 Information of meters

| Meter | Type     | Serial No. | Accuracy | Calibration frequency | Calibration date                                                                               | Validity                                                                                       |
|-------|----------|------------|----------|-----------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| M1    | ZMD405   | 85618867   | 0.5S     | Annually              | 10/10/2011<br>08/10/2012<br>07/10/2013<br>06/10/2014<br>03/10/2015<br>29/09/2016<br>26/09/2017 | 09/10/2012<br>07/10/2013<br>06/10/2014<br>05/10/2015<br>02/10/2016<br>28/09/2017<br>25/09/2018 |
| M3    | ZMD405   | 85618868   | 0.5S     |                       |                                                                                                |                                                                                                |
| M2,1  | DTSD1088 | 81101167   | 0.2S     |                       | 06/01/2012<br>04/01/2013<br>03/01/2014<br>02/01/2015<br>30/12/2015<br>26/12/2016<br>23/12/2017 | 05/01/2013<br>03/01/2014<br>02/01/2015<br>01/01/2016<br>29/12/2016<br>25/12/2017<br>22/12/2018 |
| M2,2  | DTSD1088 | 81101171   | 0.2S     |                       |                                                                                                |                                                                                                |
| M2,3  | DTSD1088 | 81101165   | 0.2S     |                       |                                                                                                |                                                                                                |
| M2,4  | DTSD1088 | 81101164   | 0.2S     |                       |                                                                                                |                                                                                                |
| M2,5  | DTSD1088 | 81101168   | 0.2S     |                       |                                                                                                |                                                                                                |
| M2,6  | DTSD1088 | 81101169   | 0.2S     |                       |                                                                                                |                                                                                                |
| M2,7  | DTSD1088 | 81101170   | 0.2S     |                       |                                                                                                |                                                                                                |
| M2,8  | DTSD1088 | 81101166   | 0.2S     |                       |                                                                                                |                                                                                                |
| M4,1  | DTSD1088 | 91102451   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,2  | DTSD1088 | 91102455   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,3  | DTSD1088 | 91102453   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,4  | DTSD1088 | 91102450   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,5  | DTSD1088 | 91102448   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,6  | DTSD1088 | 91102454   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,7  | DTSD1088 | 91102449   | 0.5S     |                       |                                                                                                |                                                                                                |
| M4,8  | DTSD1088 | 91102452   | 0.5S     |                       |                                                                                                |                                                                                                |

### 4.3 Monitoring Plan

#### Organization Structure

The figure below outlines the Project's operational and management structure. Operational and monitoring manager of the plant is responsible for the collection of the information and data required by the Monitoring Plan. The collected information is documented and sent to the VCS manager and responsible staffs of the China Resources Wind Power (Shantou) Co., Ltd. monthly. The VCS project manager is in charge of the implementation of the Monitoring Plan and report to the General Manager of the company. The General Manager of the company makes the confirmations on monitoring, calculation data and reports.

For details regarding the management structure of the monitoring plan, please refer to Figure 1.

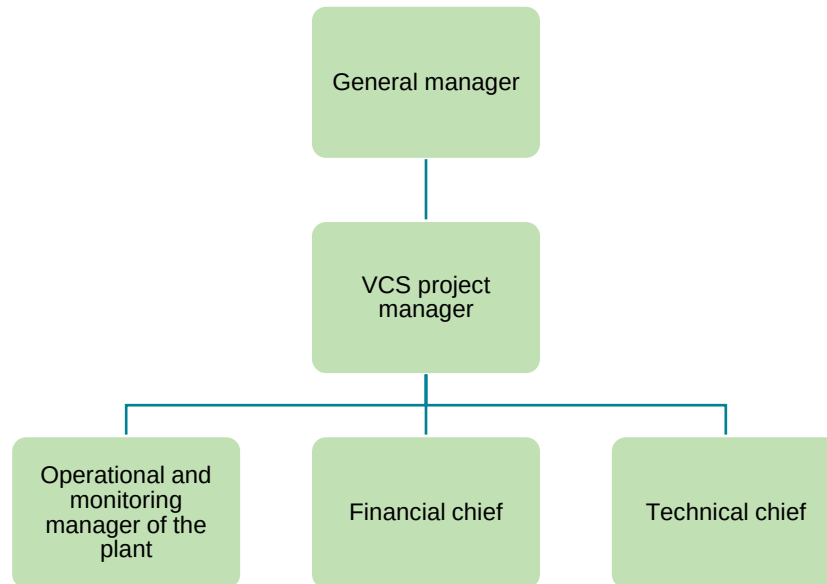


Figure 1. Management Structure of Monitoring Plan

#### Monitoring equipment and data collection procedures

Electricity supplied to ( $EG_{output,y}$ ) and purchased from ( $EG_{import,y}$ ) the grid by the Project and Nanguo Wind Farm are monitored through the bidirectional electricity energy meters installed. The main meter M1 is installed at outlet of the local 110kV substation at the project site. Another electricity energy meter M3 is installed at inlet of the 110kV local substation as the backup meter in case the error happens to the main meter M1. The accuracy level of the above meters M1 and M3 is 0.5S.

Electricity generated by Nanguo wind farm ( $EG_{Nanguooutput,y}$ ) is monitored by eight main meters (M2.1-M2,8) installed at wind turbines of Nanguo Wind Farm. The accuracy level of the main

meters is 0.2S. To ensure the electricity generated by Nanguo Wind Farm correctly measured and recorded in case that the main meters (M2,1-M2, 8) are damaged, there are eight backup meters (M4,1-M4,8) installed to monitor the  $EG_{Nanguooutput,y}$  as well. The accuracy level of the above backup meters is 0.5S.

The project proponent and the grid company record the readings of  $EG_{output,y}$  and  $EG_{import,y}$  monthly while the project proponent read and record the meter's readings of  $EG_{Nanguooutput,y}$  monthly. The cut-off time of monthly electricity data is at 24:00 on one day from 24<sup>th</sup> to the last day of each month which is determined by the grid company.

The location of the meters is as follows:

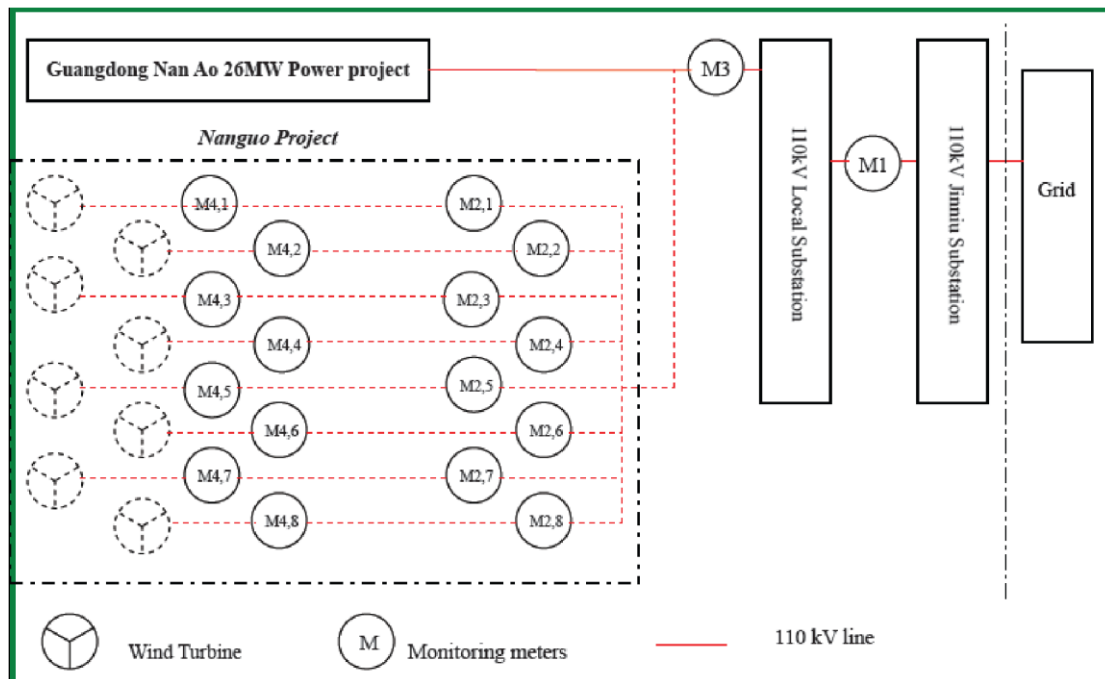


Figure 2. Monitoring points

The following formula is applied to calculate the net electricity supplied by the Project:

$$EG_y = EG_{output,y} - EG_{Nanguooutput,y} - EG_{import,y}$$

Where:

$EG_y$  = Net electricity supplied by the project in the year y.

$EG_{output,y}$  = Electricity supplied by the Project and Nanguo Wind Farm in the year y;

$EG_{Nanguooutput,y}$  = Electricity generated by Nanguo wind farm during year y;

$EG_{import,y}$  = Electricity purchased from the grid by the Project and Nanguo Wind Farm in the year y.

Of which,

$$EG_{Nanguooutput,y} = \sum_{i=1}^8 EG_{M2,i}$$

Where,

$EG_{M2,i,y}$  = Electricity generated by turbine i of Nanguo Wind Farm measured by meter (M2,i)

i = The number of turbine of Nanguo wind farm, from 1 to 8.

### Calibration of equipment and emergency procedures

The accuracy of electricity energy meter is not lower than 0.5s which meets the national requirement. The calibration frequency is annual and the calibration is carried out by a designated third party. The calibration records are provided to the project owner.

In case of that the main meter (M1) is out of service, the back-up meter (M3) will be used as monitoring of the electricity supplied to the Grid and electricity purchased from the Grid.

In case of that both the main meter (M1) and the back-up meter (M3) are out of service, the Grid Company and the project owner will jointly calculate a conservative estimation of the electricity missed. A statement will be prepared with the following information included:

- the description of the metering equipment which is out of service
- the assumptions used to estimate the electricity missed during the days of which no record could be recorded
- the estimation of the electricity missed.

The statement will be signed by both a representative of the project owner and a representative of the Grid Company.

If any error is detected, the party owning the meter shall repair, recalibrate or replace the meter giving the other party sufficient notice to allow a representative to attend during any corrective activity.

In case of that the main meters (M2,1-M2,8) of Nanguo Wind Farm are out of service, the data recorded by the meters (M4,1-M4,8) are used for calculation.

A statement will be prepared with the following information included:

- the description of the metering equipment which is out of service
- the estimation of the electricity missed.

The statement will be signed by both a representative of the project owner and a representative of Shenzhen Nanguo Energy Co., Ltd.

During this monitoring period, no non-conformities were identified.

### Data Management and QA/QC

Record keeping is the most important exercise in relation to the monitoring process. Below follows an outline of how project related records would be managed.

Overall responsibility for monitoring of GHG emissions reduction rests with the VCS responsible person of the project. The VCS manual sets out the procedures for tracking information from the primary source to the end-data calculations in paper document format. It is the responsibility of the project proponent to provide additional necessary data and information for verification requirements of respective VVB. Physical documentation such as paper-based records, diagrams and environmental assessment are collated in a central place, together with this monitoring plan. All paper-based information is stored by the project proponent and kept at least one copy and all data or records will be archived at least 2 years after the end of the final crediting period

## 5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

### 5.1 Baseline Emissions

As per the registered PD, the baseline emission of the project is calculated as below:

$$BE_y = EG_y * EF_y$$

Table 5 Baseline emissions calculation

| Year                             | EG <sub>y</sub>    | EF <sub>y</sub>        | BE <sub>y</sub>    |
|----------------------------------|--------------------|------------------------|--------------------|
|                                  | MWh                | tCO <sub>2</sub> e/MWh | tCO <sub>2</sub> e |
| 2012<br>(29/09/2012- 31/12/2012) | 19,480.041         | 0.9276                 | 18,069             |
| 2013<br>(01/01/2013- 28/12/2013) | 57,528.909         | 0.9276                 | 53,363             |
| 2014<br>(29/12/2013- 31/12/2014) | 52,401.184         | 0.9276                 | 48,607             |
| 2015<br>(01/01/2015- 31/12/2015) | 53,747.669         | 0.9276                 | 49,856             |
| 2016<br>(01/01/2016- 28/12/2016) | 53,109.896         | 0.9276                 | 49,264             |
| 2017<br>(29/12/2016- 31/12/2017) | 60,726.988         | 0.9276                 | 56,330             |
| <b>Total</b>                     | <b>296,994.687</b> | <b>/</b>               | <b>275,489</b>     |

The monitored monthly electricity data is shown in the table below:

Table 6 Monthly monitored electricity data and calculation of EG<sub>y</sub> (unit: MWh)

| Start         | End        | EG <sub>output,y</sub> from meter readings of M1 | EG <sub>output,y</sub> from ETNs | Conservative values of EG <sub>output,y</sub> | EG <sub>Nanguooutput,y</sub> measured by M2,1- M2,8 | EG <sub>import,y</sub> from meter readings of M1 | EG <sub>import,y</sub> from ETNs | Conservative values of EG <sub>import,y</sub> | EG <sub>y</sub> = EG <sub>output,y</sub> - EG <sub>Nanguooutput,y</sub> - EG <sub>import,y</sub> |
|---------------|------------|--------------------------------------------------|----------------------------------|-----------------------------------------------|-----------------------------------------------------|--------------------------------------------------|----------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------|
| 29/09/2012    | 28/10/2012 | 5,724.400                                        | 5,682.910                        | 5,682.910                                     | 1,324.800                                           | 30.800                                           | 30.800                           | 30.800                                        | 4327.310                                                                                         |
| 29/10/2012    | 30/11/2012 | 8,342.400                                        | 8,283.592                        | 8,283.592                                     | 1,720.680                                           | 4.400                                            | 4.400                            | 4.400                                         | 6558.512                                                                                         |
| 01/12/2012    | 31/12/2012 | 10,516.000                                       | 10,438.939                       | 10,438.939                                    | 1,840.320                                           | 4.400                                            | 4.400                            | 4.400                                         | 8594.219                                                                                         |
| Subtotal 2012 |            | -                                                | -                                | 24,405.441                                    | 4,885.800                                           | -                                                | -                                | 39.600                                        | 19,480.041                                                                                       |
| 01/01/2013    | 28/01/2013 | 8,663.600                                        | 8,596.104                        | 8,596.104                                     | 1,610.520                                           | 8.800                                            | 8.800                            | 8.800                                         | 6976.784                                                                                         |
| 29/01/2013    | 26/02/2013 | 8,100.400                                        | 8,032.517                        | 8,032.517                                     | 1,298.040                                           | 4.400                                            | 4.400                            | 4.400                                         | 6730.077                                                                                         |
| 27/02/2013    | 28/03/2013 | 5,183.200                                        | 5,143.519                        | 5,143.519                                     | 1,017.120                                           | 13.200                                           | 13.200                           | 13.200                                        | 4113.199                                                                                         |
| 29/03/2013    | 27/04/2013 | 6,978.400                                        | 6,920.604                        | 6,920.604                                     | 1,565.400                                           | 17.600                                           | 17.600                           | 17.600                                        | 5337.604                                                                                         |
| 28/04/2013    | 28/05/2013 | 4,747.600                                        | 4,712.207                        | 4,712.207                                     | 981.000                                             | 13.200                                           | 13.200                           | 13.200                                        | 3718.007                                                                                         |
| 29/05/2013    | 27/06/2013 | 3,462.800                                        | 3,434.394                        | 3,434.394                                     | 722.880                                             | 17.600                                           | 17.600                           | 17.600                                        | 2693.914                                                                                         |
| 28/06/2013    | 28/07/2013 | 2,178.000                                        | 2,159.803                        | 2,159.803                                     | 381.960                                             | 17.600                                           | 17.600                           | 17.600                                        | 1760.243                                                                                         |
| 29/07/2013    | 28/08/2013 | 2,094.400                                        | 2,079.282                        | 2,079.282                                     | 363.240                                             | 26.400                                           | 26.400                           | 26.400                                        | 1689.642                                                                                         |
| 29/08/2013    | 28/09/2013 | 6,833.200                                        | 6,781.371                        | 6,781.371                                     | 1,695.960                                           | 35.200                                           | 35.200                           | 35.200                                        | 5050.211                                                                                         |
| 29/09/2013    | 28/10/2013 | 4,989.600                                        | 4,950.161                        | 4,950.161                                     | 1,018.800                                           | 8.800                                            | 8.800                            | 8.800                                         | 3922.561                                                                                         |
| 29/10/2013    | 28/11/2013 | 10,986.800                                       | 10,906.657                       | 10,906.657                                    | 2,650.920                                           | 4.400                                            | 4.400                            | 4.400                                         | 8251.337                                                                                         |
| 29/11/2013    | 28/12/2013 | 9,468.800                                        | 9,402.090                        | 9,402.090                                     | 2,112.360                                           | 4.400                                            | 4.400                            | 4.400                                         | 7285.330                                                                                         |
| Subtotal 2013 |            | -                                                | -                                | 73,118.709                                    | 15,418.200                                          | -                                                | -                                | 171.600                                       | 57,528.909                                                                                       |
| 29/12/2013    | 25/01/2014 | 6,894.800                                        | 6,846.175                        | 6,846.175                                     | 1,112.520                                           | 4.400                                            | 4.400                            | 4.400                                         | 5729.255                                                                                         |
| 26/01/2014    | 28/02/2014 | 6,991.600                                        | 6,921.804                        | 6,921.804                                     | 1,503.000                                           | 8.800                                            | 8.800                            | 8.800                                         | 5410.004                                                                                         |
| 01/03/2014    | 28/03/2014 | 6,529.600                                        | 6,456.314                        | 6,456.314                                     | 1,107.600                                           | 17.600                                           | 17.600                           | 17.600                                        | 5331.114                                                                                         |
| 29/03/2014    | 28/04/2014 | 4,589.200                                        | 4,547.780                        | 4,547.780                                     | 717.240                                             | 17.600                                           | 17.600                           | 17.600                                        | 3812.940                                                                                         |
| 29/04/2014    | 28/05/2014 | 4,483.600                                        | 4,436.411                        | 4,436.411                                     | 870.240                                             | 17.600                                           | 17.600                           | 17.600                                        | 3548.571                                                                                         |
| 29/05/2014    | 28/06/2014 | 5,042.400                                        | 4,970.242                        | 4,970.242                                     | 851.400                                             | 17.600                                           | 17.600                           | 17.600                                        | 4101.242                                                                                         |

|               |            |            |            |            |            |        |        |         |            |
|---------------|------------|------------|------------|------------|------------|--------|--------|---------|------------|
| 29/06/2014    | 28/07/2014 | 1,804.000  | 1,783.773  | 1,783.773  | 455.640    | 17.600 | 17.600 | 17.600  | 1310.533   |
| 29/07/2014    | 28/08/2014 | 1,592.800  | 1,574.411  | 1,574.411  | 297.000    | 30.800 | 30.800 | 30.800  | 1246.611   |
| 29/08/2014    | 28/09/2014 | 1,848.000  | 1,828.393  | 1,828.393  | 363.120    | 39.600 | 39.600 | 39.600  | 1425.673   |
| 29/09/2014    | 28/10/2014 | 6,674.800  | 6,597.496  | 6,597.496  | 1,116.240  | 35.200 | 35.200 | 35.200  | 5446.056   |
| 29/10/2014    | 28/11/2014 | 8,016.800  | 7,925.614  | 7,925.614  | 1,297.680  | 13.200 | 13.200 | 13.200  | 6614.734   |
| 29/11/2014    | 31/12/2014 | 10,472.000 | 10,350.411 | 10,350.411 | 1,921.560  | 4.400  | 4.400  | 4.400   | 8424.451   |
| Subtotal 2014 |            | -          | -          | 64,238.824 | 11,613.240 | -      | -      | 224.400 | 52,401.184 |
| 01/01/2015    | 28/01/2015 | 9,512.800  | 9,390.035  | 9,390.035  | 1,976.280  | 4.400  | 4.400  | 4.400   | 7409.355   |
| 29/01/2015    | 26/02/2015 | 6,490.000  | 6,424.003  | 6,424.003  | 1,196.880  | 8.800  | 8.800  | 8.800   | 5218.323   |
| 27/02/2015    | 28/03/2015 | 7,656.000  | 7,578.747  | 7,578.747  | 1,659.600  | 13.200 | 13.200 | 13.200  | 5905.947   |
| 29/03/2015    | 28/04/2015 | 5,460.400  | 5,398.731  | 5,398.731  | 882.360    | 22.000 | 22.000 | 22.000  | 4494.371   |
| 29/04/2015    | 28/05/2015 | 3,669.600  | 3,620.980  | 3,620.980  | 847.800    | 22.000 | 22.000 | 22.000  | 2751.180   |
| 29/05/2015    | 28/06/2015 | 3,022.800  | 2,989.352  | 2,989.352  | 567.240    | 13.200 | 13.200 | 13.200  | 2408.912   |
| 29/06/2015    | 28/07/2015 | 4,382.400  | 4,336.217  | 4,336.217  | 846.240    | 13.200 | 13.200 | 13.200  | 3476.777   |
| 29/07/2015    | 28/08/2015 | 1,927.200  | 1,901.555  | 1,901.555  | 375.840    | 30.800 | 30.800 | 30.800  | 1494.915   |
| 29/08/2015    | 28/09/2015 | 3,190.000  | 3,158.615  | 3,158.615  | 807.480    | 35.200 | 35.200 | 35.200  | 2315.935   |
| 29/09/2015    | 29/10/2015 | 6,498.800  | 6,420.926  | 6,420.926  | 1,064.520  | 30.800 | 30.800 | 30.800  | 5325.606   |
| 30/10/2015    | 29/11/2015 | 7,902.400  | 7,817.666  | 7,817.666  | 1,292.280  | 8.800  | 8.800  | 8.800   | 6516.586   |
| 30/11/2015    | 31/12/2015 | 8,452.400  | 8,367.762  | 8,367.762  | 1,924.800  | 13.200 | 13.200 | 13.200  | 6429.762   |
| Subtotal 2015 |            | -          | -          | 67,404.589 | 13,441.320 | -      | -      | 215.600 | 53,747.669 |
| 01/01/2016    | 29/01/2016 | 8,307.200  | 8,222.952  | 8,222.952  | 2,011.440  | 13.200 | 13.200 | 13.200  | 6198.312   |
| 30/01/2016    | 27/02/2016 | 6,912.400  | 6,828.152  | 6,828.152  | 1,140.960  | 4.400  | 4.400  | 4.400   | 5682.792   |
| 28/02/2016    | 29/03/2016 | 8,038.800  | 7,954.552  | 7,954.552  | 1,277.280  | 8.800  | 8.800  | 8.800   | 6668.472   |
| 30/03/2016    | 28/04/2016 | 2,481.600  | 2,397.352  | 2,397.352  | 404.520    | 13.200 | 13.200 | 13.200  | 1979.632   |
| 29/04/2016    | 30/05/2016 | 4,936.800  | 4,852.552  | 4,852.552  | 980.280    | 26.400 | 26.400 | 26.400  | 3845.872   |
| 31/05/2016    | 29/06/2016 | 2,864.400  | 2,780.152  | 2,780.152  | 606.000    | 17.600 | 17.600 | 17.600  | 2156.552   |
| 30/06/2016    | 29/07/2016 | 1,883.200  | 1,798.952  | 1,798.952  | 459.600    | 26.400 | 26.400 | 26.400  | 1312.952   |
| 30/07/2016    | 30/08/2016 | 2,569.600  | 2,485.352  | 2,485.352  | 643.560    | 26.400 | 26.400 | 26.400  | 1815.392   |
| 31/08/2016    | 29/09/2016 | 4,228.400  | 4,228.400  | 4,228.400  | 963.120    | 44.000 | 44.000 | 44.000  | 3221.280   |
| 30/09/2016    | 30/10/2016 | 7,722.000  | 7,722.000  | 7,722.000  | 1,202.640  | 26.400 | 26.400 | 26.400  | 6492.960   |
| 31/10/2016    | 28/11/2016 | 7,920.000  | 7,840.800  | 7,840.800  | 1,611.720  | 17.600 | 17.600 | 17.600  | 6211.480   |

|               |            |            |            |             |            |        |        |           |             |
|---------------|------------|------------|------------|-------------|------------|--------|--------|-----------|-------------|
| 29/11/2016    | 28/12/2016 | 9,944.000  | 9,844.560  | 9,844.560   | 2,307.600  | 12.760 | 12.760 | 12.760    | 7524.200    |
| Subtotal 2016 |            | -          | -          | 66,955.776  | 13,608.720 | -      | -      | 237.160   | 53,109.896  |
| 29/12/2016    | 24/01/2017 | 10,401.600 | 10,297.584 | 10,297.584  | 2,108.400  | 4.840  | 4.840  | 4.840     | 8184.344    |
| 25/01/2017    | 26/02/2017 | 9,640.400  | 9,543.996  | 9,543.996   | 1,636.320  | 4.400  | 4.400  | 4.400     | 7903.276    |
| 27/02/2017    | 29/03/2017 | 7,937.600  | 7,858.224  | 7,858.224   | 1,395.840  | 13.200 | 13.200 | 13.200    | 6449.184    |
| 30/03/2017    | 28/04/2017 | 4,690.400  | 4,643.496  | 4,643.496   | 910.560    | 8.800  | 8.800  | 8.800     | 3724.136    |
| 29/04/2017    | 29/05/2017 | 3,898.400  | 3,859.416  | 3,859.416   | 814.320    | 26.400 | 26.400 | 26.400    | 3018.696    |
| 30/05/2017    | 28/06/2017 | 3,942.400  | 3,902.976  | 3,902.976   | 656.040    | 22.000 | 22.000 | 22.000    | 3224.936    |
| 29/06/2017    | 29/07/2017 | 1,161.600  | 1,149.984  | 1,149.984   | 222.600    | 22.000 | 22.000 | 22.000    | 905.384     |
| 30/07/2017    | 29/08/2017 | 3,137.200  | 3,105.828  | 3,105.828   | 656.640    | 39.600 | 39.600 | 39.600    | 2409.588    |
| 30/08/2017    | 28/09/2017 | 2,578.400  | 2,552.616  | 2,552.616   | 464.880    | 26.400 | 26.400 | 26.400    | 2061.336    |
| 29/09/2017    | 30/10/2017 | 7,876.000  | 7,797.240  | 7,797.240   | 1,829.640  | 39.600 | 39.600 | 39.600    | 5928.000    |
| 31/10/2017    | 28/11/2017 | 11,127.600 | 11,016.324 | 11,016.324  | 2,052.960  | 8.800  | 8.800  | 8.800     | 8954.564    |
| 31/10/2017    | 29/12/2017 | 9,653.600  | 9,557.064  | 9,557.064   | 1,584.720  | 4.400  | 4.400  | 4.400     | 7967.944    |
| 30/12/2017    | 31/12/2017 | 0.000      | 0.000      | 0.000       | 0.000      | 4.400  | 4.400  | 4.400     | -4.400      |
| Subtotal 2017 |            | -          | -          | 75,284.748  | 14,332.920 | -      | -      | 224.840   | 60,726.988  |
| Total         |            | -          | -          | 371,408.087 | 73,300.200 | -      | -      | 1,113.200 | 296,994.687 |

1. Since the cut-off time of December for each year is not always on the last day i.e. the 31<sup>st</sup> Dec, the electricity data for 29/12/2013-31/12/2013 and 29/12/2016-31/12/2016 is settled in Jan 2014 and Jan 2017 respectively, therefore, the ERs generated for 29/12/2013-31/12/2013 and 29/12/2016-31/12/2016 are included in the vintage 2014 and 2017 respectively.
2. As the electricity data is monthly recorded and the ETNs are issued monthly, there are no separate reading records nor ETNs for the last two days of this monitoring period, i.e. 30<sup>th</sup>- 31<sup>st</sup> Dec 2017. To be conservative, electricity exported by project ( $EG_{output,y} - EG_{Nanguoooutput,y}$ ) for those 2 days is considered as 0 while the whole month's (30/12/2017-28/01/2018) values of  $EG_{import,y}$  are applied for ERs calculations.

## 5.2 Project Emissions

As per the methodology and the registered PD, the project emission is 0.

## 5.3 Leakage

As per the methodology and the registered PD, the leakage of the project is not considered.

## 5.4 Net GHG Emission Reductions and Removals

| Year                             | Baseline emissions or removals (tCO <sub>2</sub> e) | Project emissions or removals (tCO <sub>2</sub> e) | Leakage emissions (tCO <sub>2</sub> e) | Net GHG emission reductions or removals (tCO <sub>2</sub> e) |
|----------------------------------|-----------------------------------------------------|----------------------------------------------------|----------------------------------------|--------------------------------------------------------------|
| 2012<br>(29/09/2012- 31/12/2012) | 18,069                                              | 0                                                  | 0                                      | 18,069                                                       |
| 2013<br>(01/01/2013- 28/12/2013) | 53,363                                              | 0                                                  | 0                                      | 53,363                                                       |
| 2014<br>(29/12/2013- 31/12/2014) | 48,607                                              | 0                                                  | 0                                      | 48,607                                                       |
| 2015<br>(01/01/2015- 31/12/2015) | 49,856                                              | 0                                                  | 0                                      | 49,856                                                       |
| 2016<br>(01/01/2016- 28/12/2016) | 49,264                                              | 0                                                  | 0                                      | 49,264                                                       |
| 2017<br>(29/12/2016- 31/12/2017) | 56,330                                              | 0                                                  | 0                                      | 56,330                                                       |
| <b>Total</b>                     | <b>275,489</b>                                      | <b>0</b>                                           | <b>0</b>                               | <b>275,489</b>                                               |

**Comparison of the actual emission reductions with the estimated values of this monitoring period is analysed as follows:**

Annual estimated emission reductions are 53,425 tCO<sub>2</sub>e

Total days during this monitoring period are 1,920 days

Calculated estimation of the emission reductions:  $53,425 * 1920 / 365 = 281,030$  tCO<sub>2</sub>e

The actual emission reductions achieved during this monitoring period are 1.97% lower than the estimated ex-ante amount. The decrease in the actual emission reductions result from fluctuation in wind power resources during this monitoring period which is a reasonable range.

# APPENDIX 1: <SUPPORTING EVIDENCE>

## **Evidence for SDG 7 and SDG 13:**

The net electricity supplied to the grid is 296,994.687 MWh and the emission reductions generated are 275,489 tCO<sub>2e</sub> during this monitoring period (29/09/2012-31/12/2017) and the detailed calculation process have been displayed in Section 5 of the MR.

## **Evidence for SDG 8:**

The evidence for SDG 8 including the number of employees and annual average income is submitted as a separate document to Verra due to confidentiality requirement of the project proponent.

This is the first time to report sustainable development contributions of the project. Therefore, the information for the project lifetime and the current monitoring period are the same as per VCS requirement.